

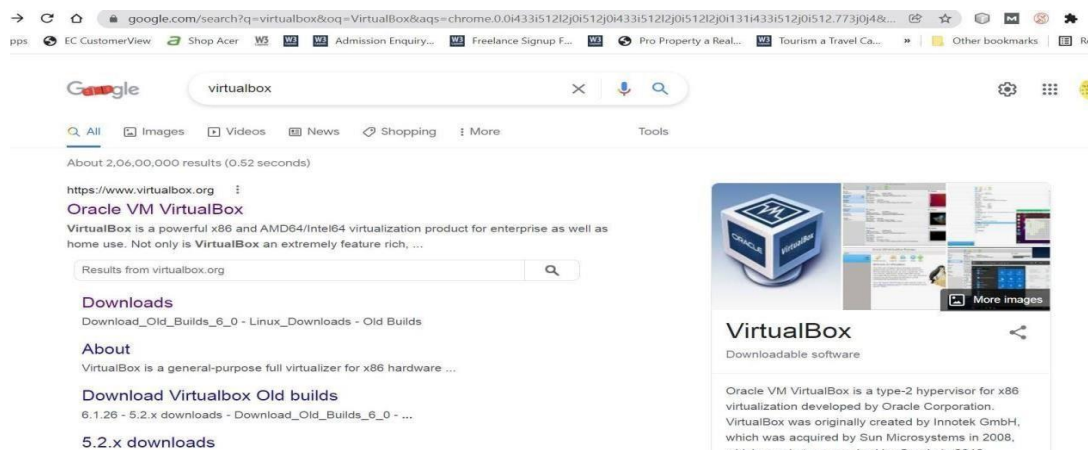
Course Code: CAP 457	Course Instructor: Dr. Apash Roy
Course Title: Big Data-Lab	Roll Number: RD2110B61
Name: Atul Kumar	Registration Number: 12102801
Assignment: 03	Page Number: 01 Total Pages: 21

Question: Prepare a Lab file with steps and screenshots for all tasks done till date.

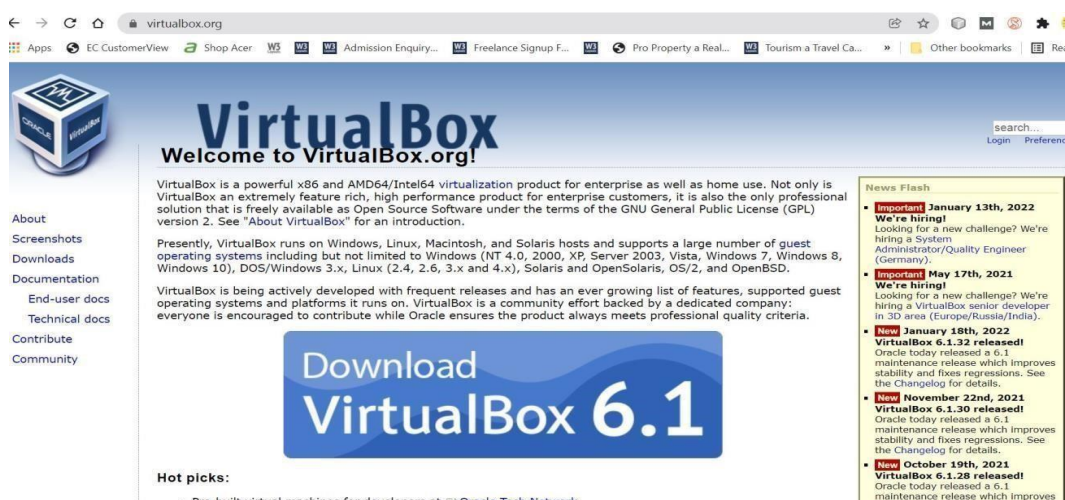
Answer: -

Task 1: STEPS TO INSTALL ORACLE VIRTUALBOX.

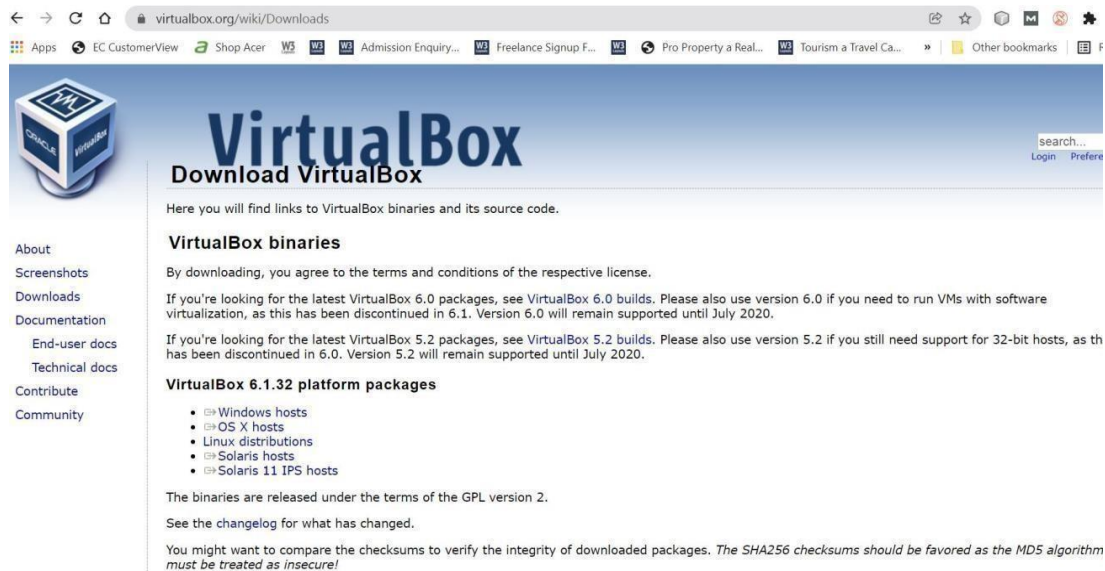
Step 1: First you have to open an internet browser then search for virtual box. You will find a link www.virtualbox.org and you have to open it by clicking on it.



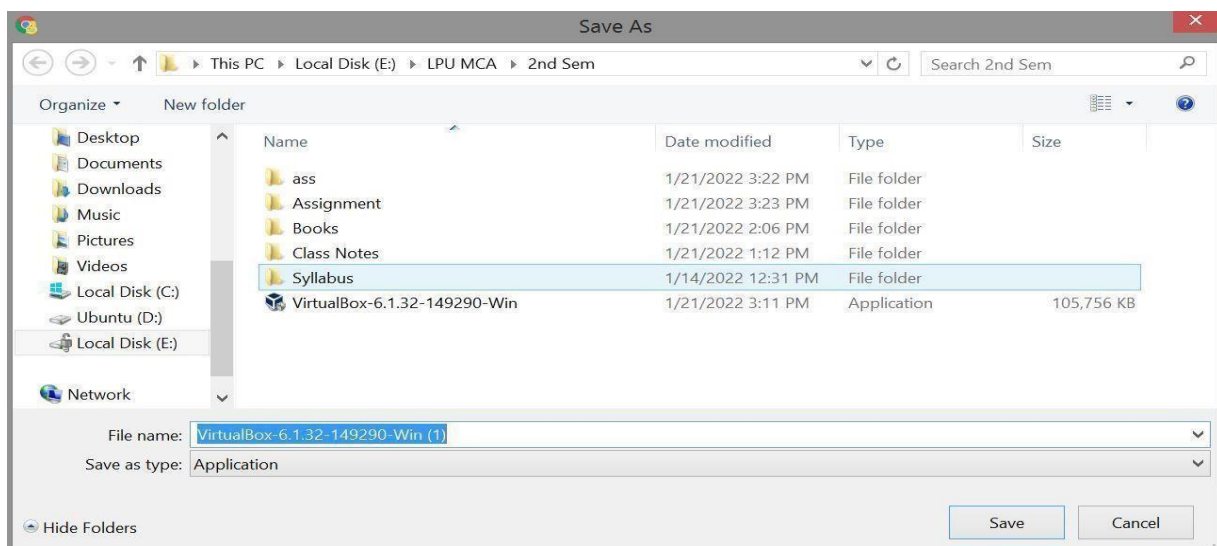
Step 2: After opening the link you have to click on the Download VirtualBox 6.1



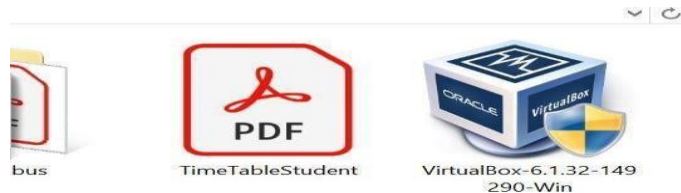
Step 3: After Clicking on download button, it will redirect you to another page where it will ask you for which kind of operating system you want on virtual box (in my case I am using windows so I clicked on windows)



Step 4: After clicking on your specific operating system name, you will receive a pop up in that you have to define where you want to download your virtual box you have to specify folder in drive.



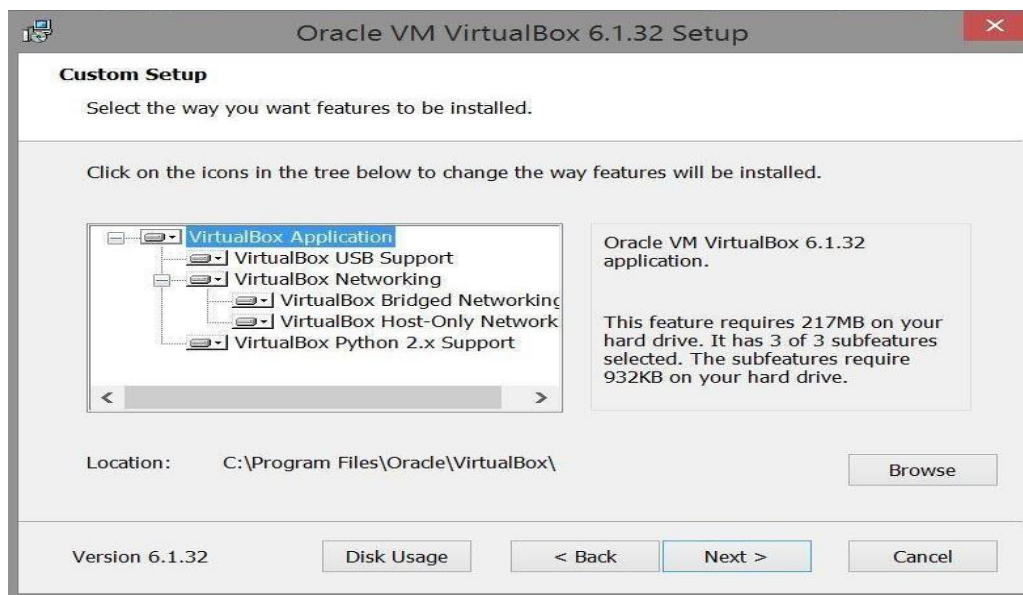
Step 5: After downloading the file you have to go to that particular folder where you downloaded your virtual box and double click on it for starting the installing process.



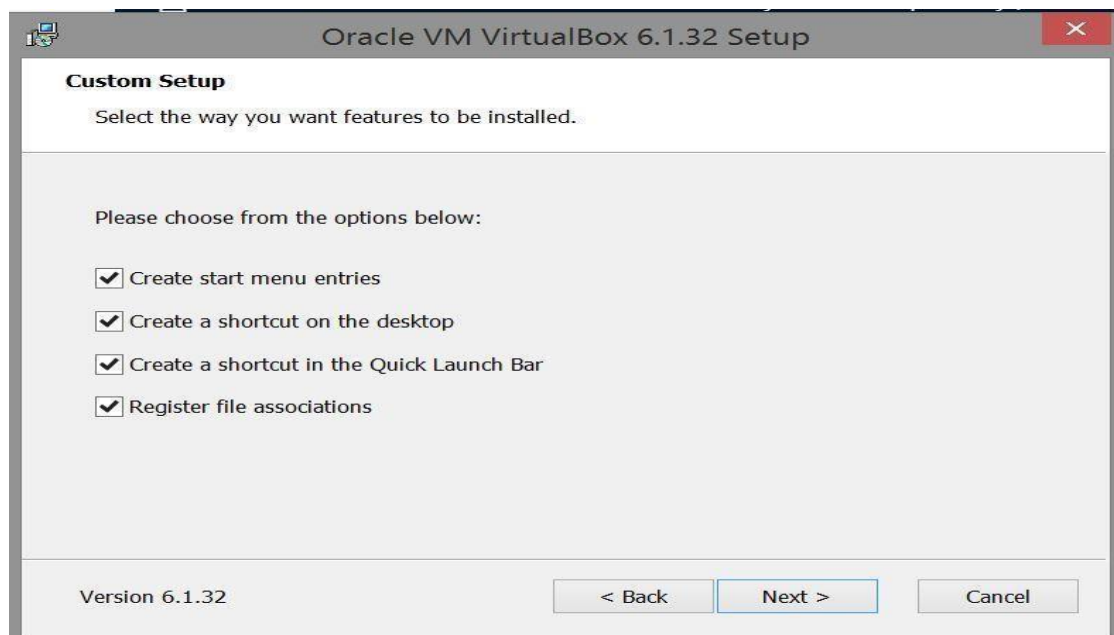
Step 6: When you double clicked this will be the first window that would be open now you have to click on next button.



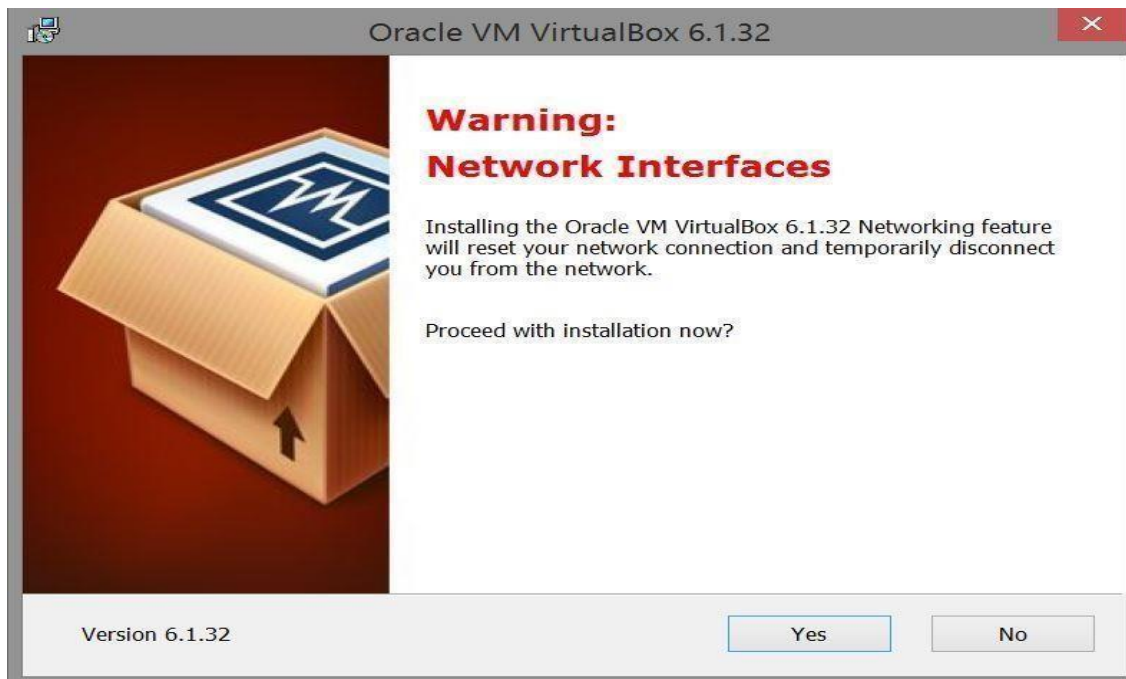
Step 7: After pressing next it will ask you to specify where your software to be installed. Specify the folder in drive and click on next.



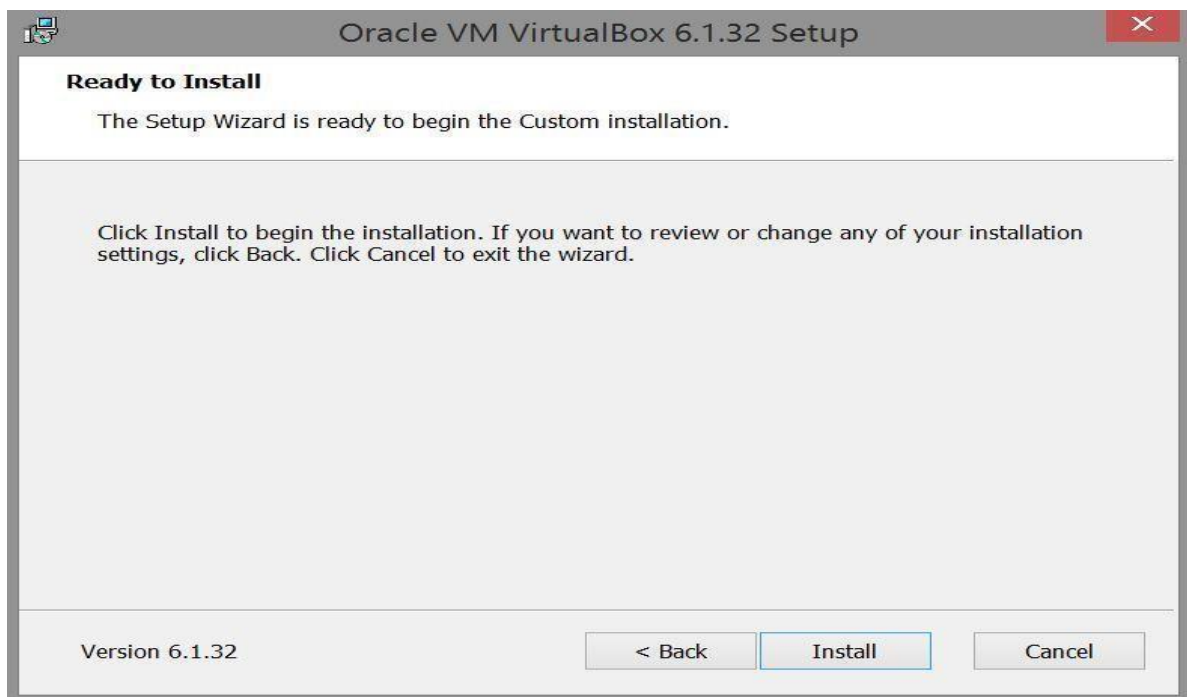
Step 8: In this window it will ask for creating shortcuts to launch the Software. Tick the boxes from where you want to access the software and click on next.



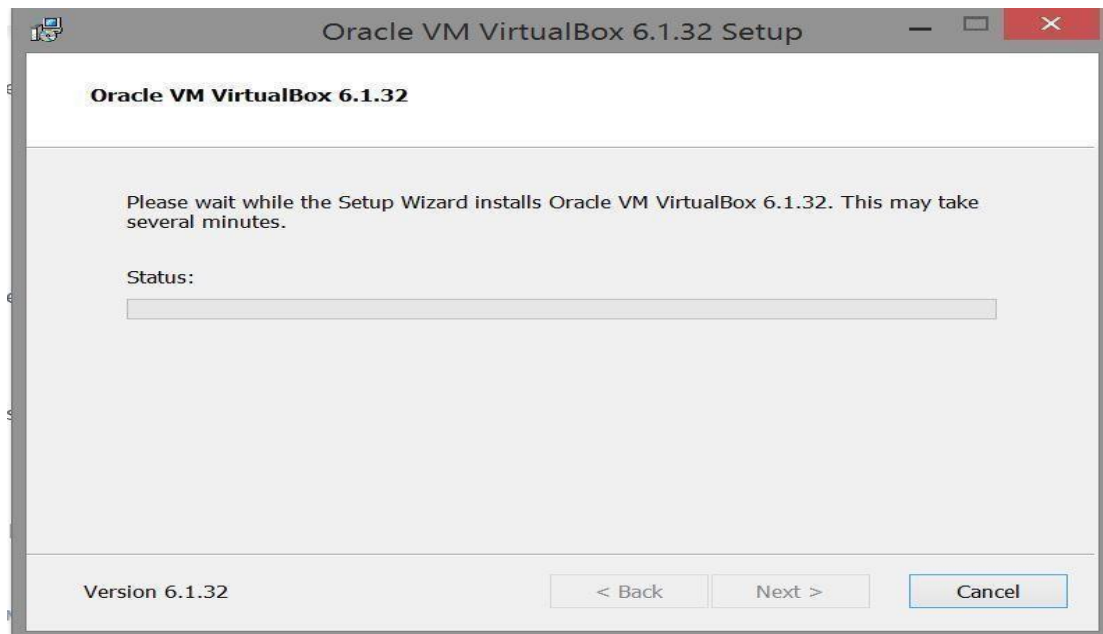
Step 9: In this window it will warn you that you may be disconnected for a while, while you are installing virtual box. Click on yes otherwise it will terminate all the processes you have done earlier.



Step 10: It will be giving you hint that now software is ready for installation process click on install.



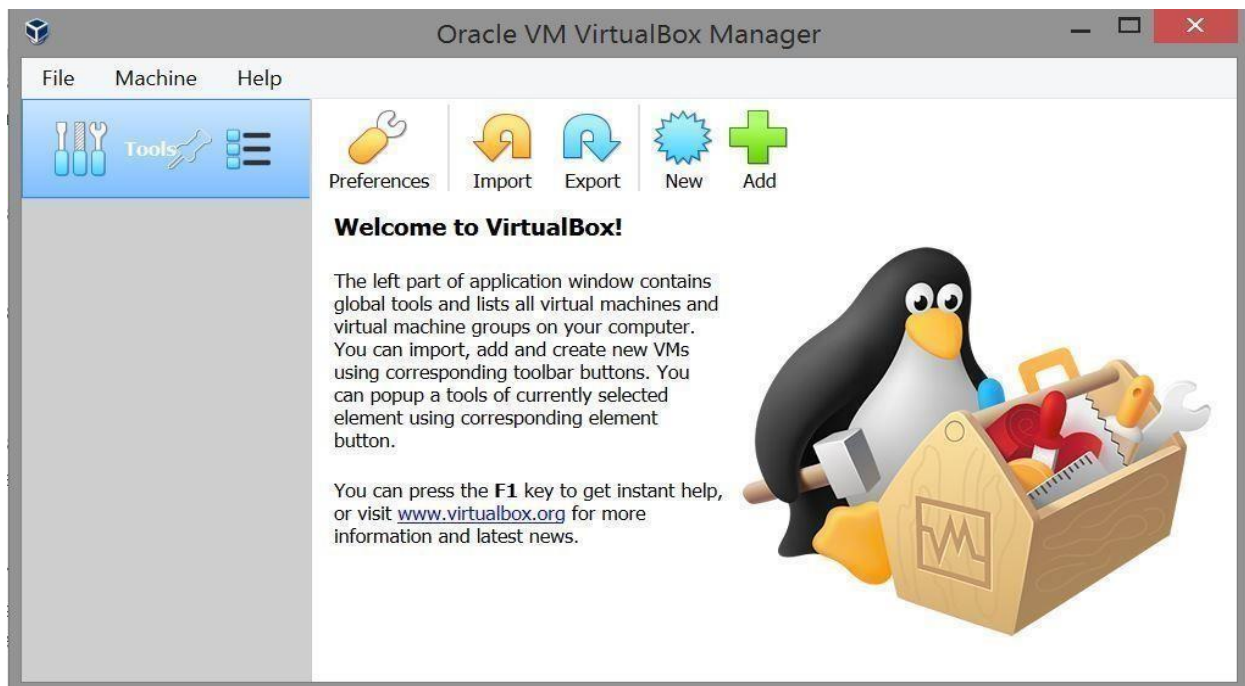
Step 11: While software is installing this window will appear, you can see the status of installation process.



Step 12: Now your Installation has completed you can finish your installation process by clicking on finish and it will start the software.

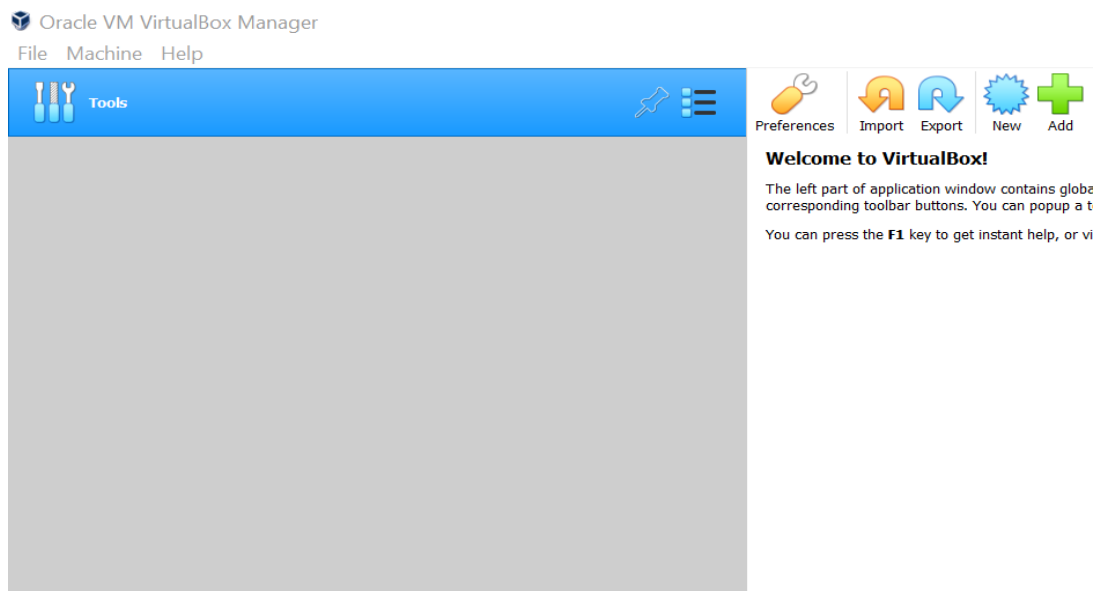


Step 13: It is the first window of your Virtual box from where you can create or access any virtual machine.

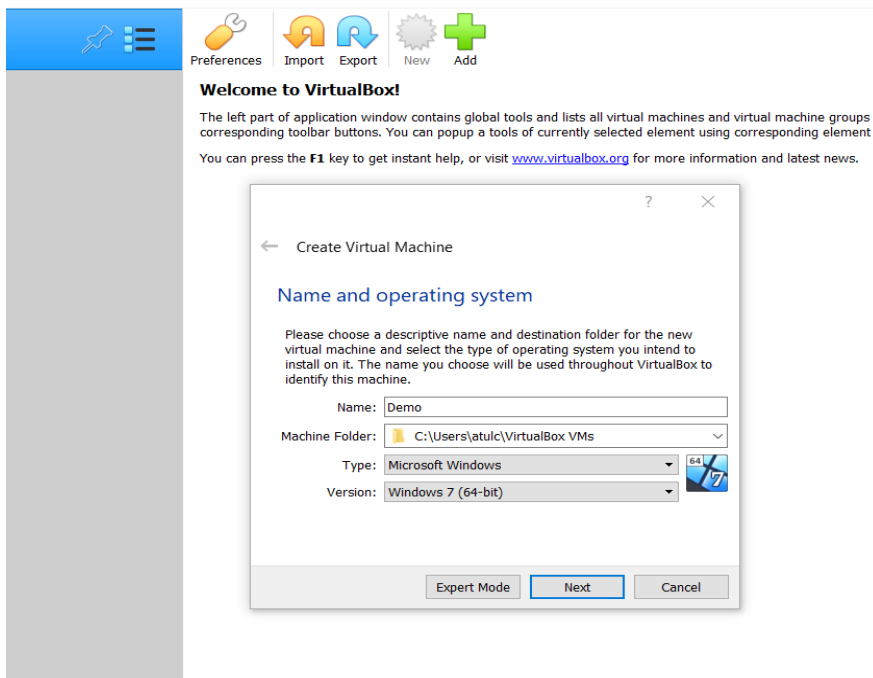


Task 2: STEPS TO CREATE VIRTUAL MACHINE.

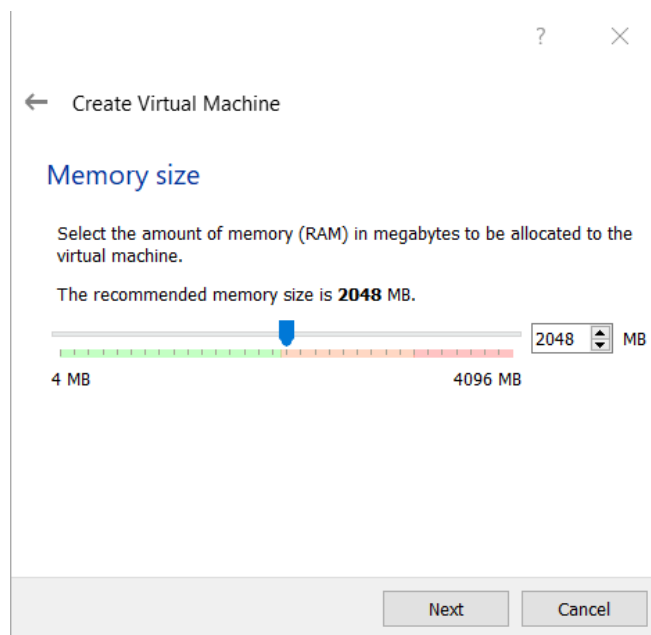
Step 1: Open Oracle VM VirtualBox then click on new option.



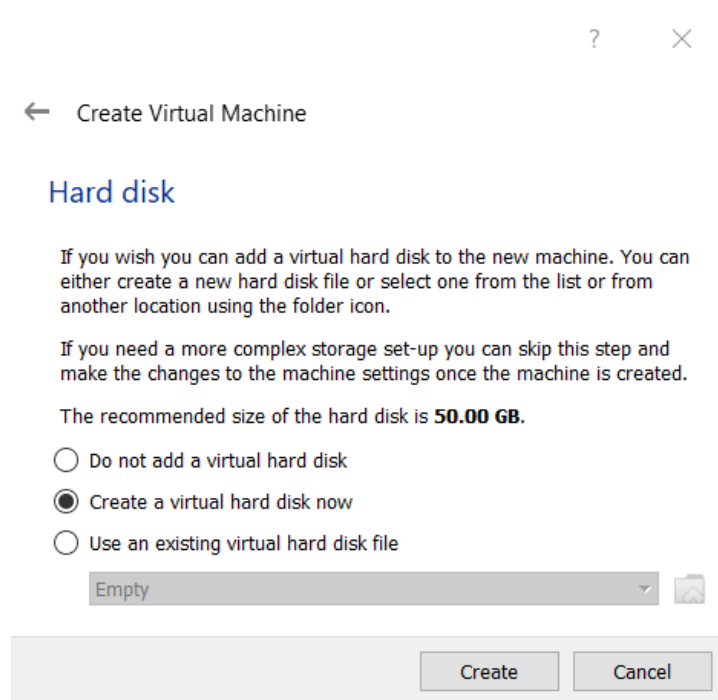
Step 2: After opening now type the name of your machine and choose operating system and its version then click on next.



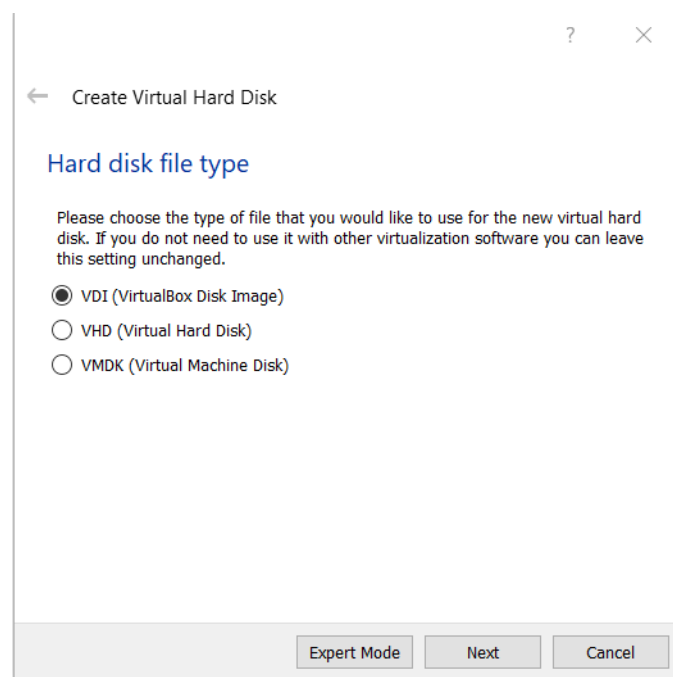
Step 3: Now you have to set memory size for the machine and then click on next.



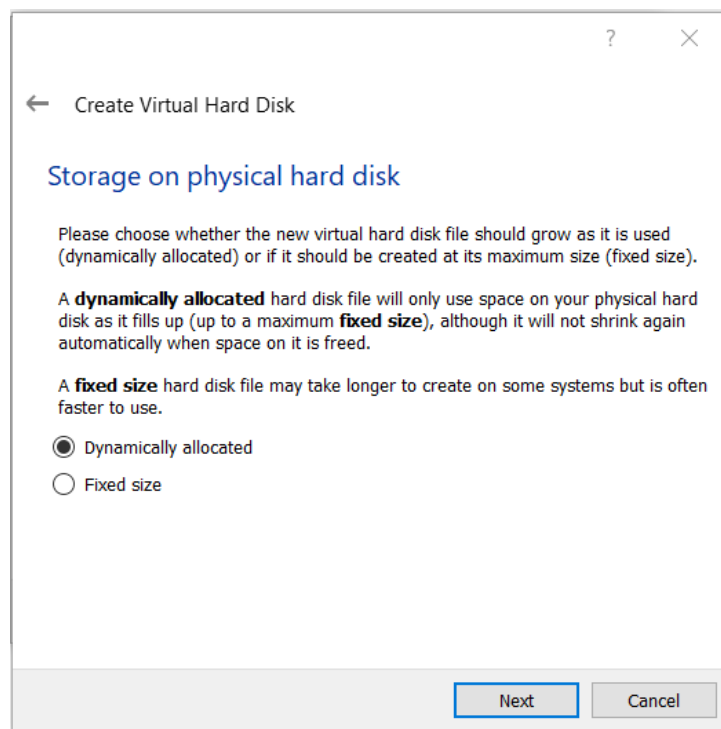
Step 4: Now for creating a virtual hard disk, click on the second option (Create a virtual hard disk now) and then click on create.



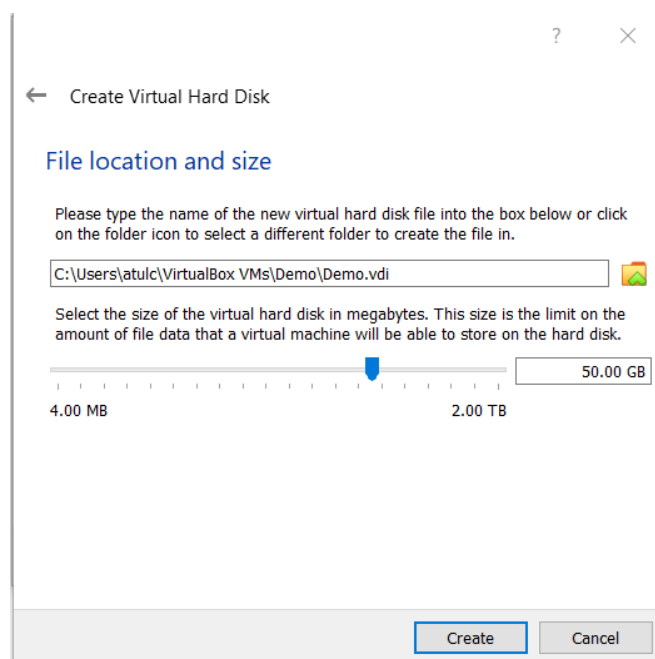
Step 5: After creating hard disk now select first option (VDI- VirtualBox Disk Image) for hard disk file type then click on next.



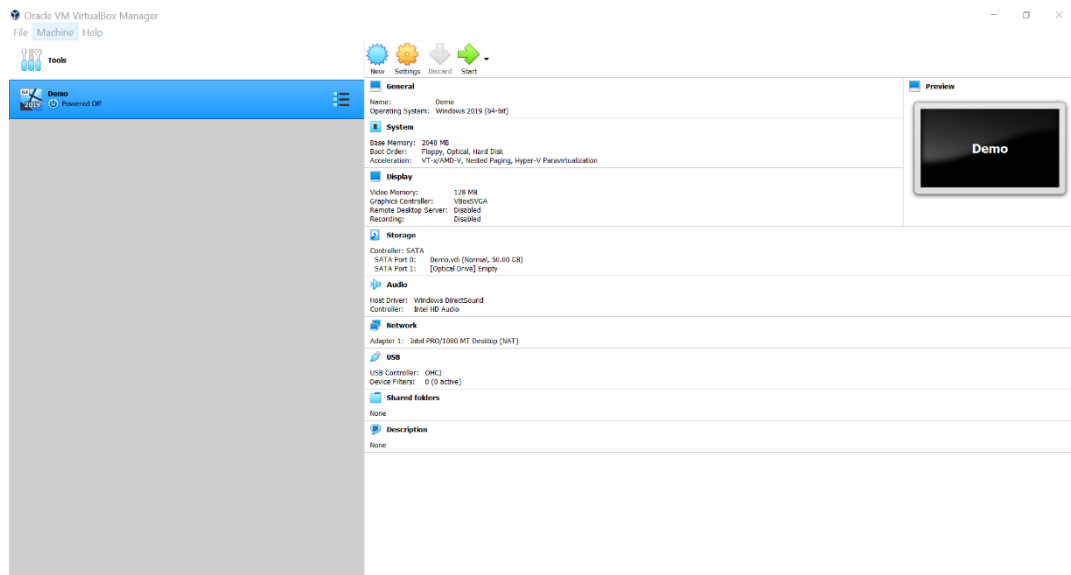
Step 6: Now select first option (Dynamically allocated) for storage on physical hard disk and then click on next.



Step 7: Now you have to select file location where you want to store your files and set size for virtual hard disk then click on create.



Step 8: Now your Virtual Machine is created and ready to use.

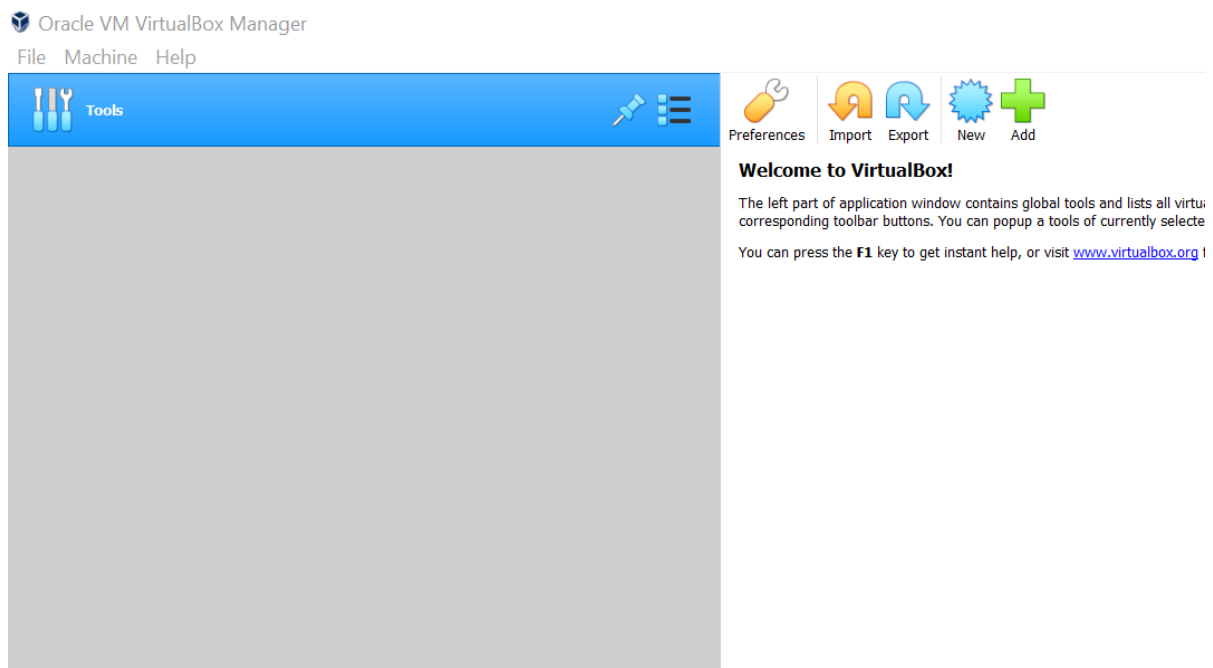


Task 3: STEPS TO INSTALL CLOUDERA.

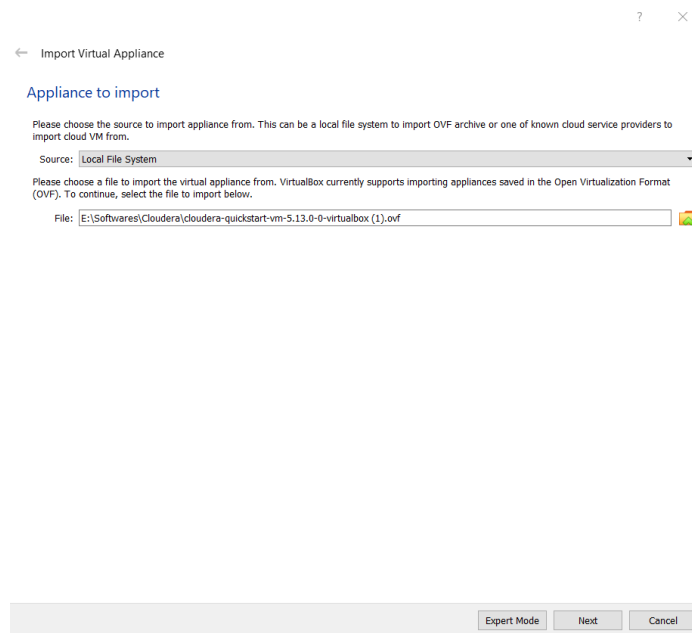
Step 1: First we have to download both the files: -

1. <https://drive.google.com/file/d/1QGFGWnFva7bQA4mazKPaejnqljkhAXEY/view?usp=sharing>
2. <https://drive.google.com/file/d/1wCGGUmgO3J3K4x83LAGdFJuVBg3br3bP/view?usp=sharing>

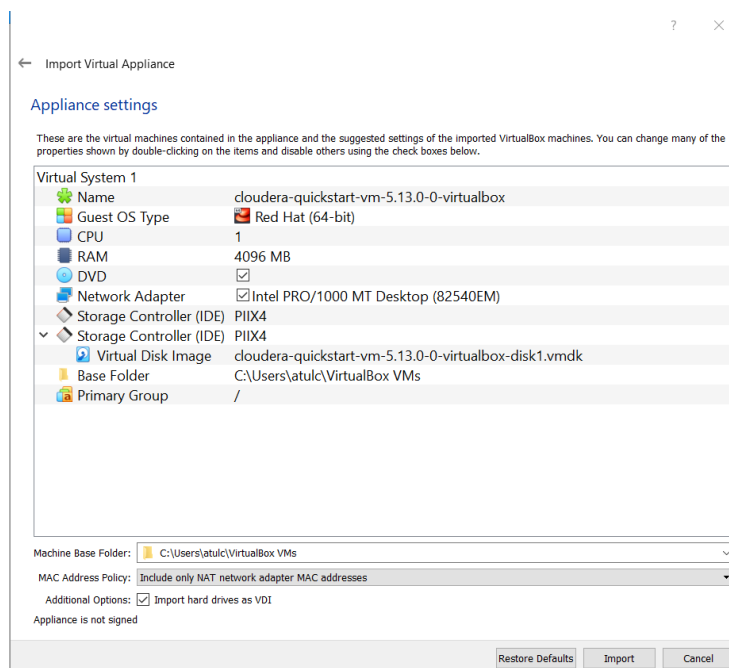
Step 2: Open Oracle VirtualBox and then click on the Import option.



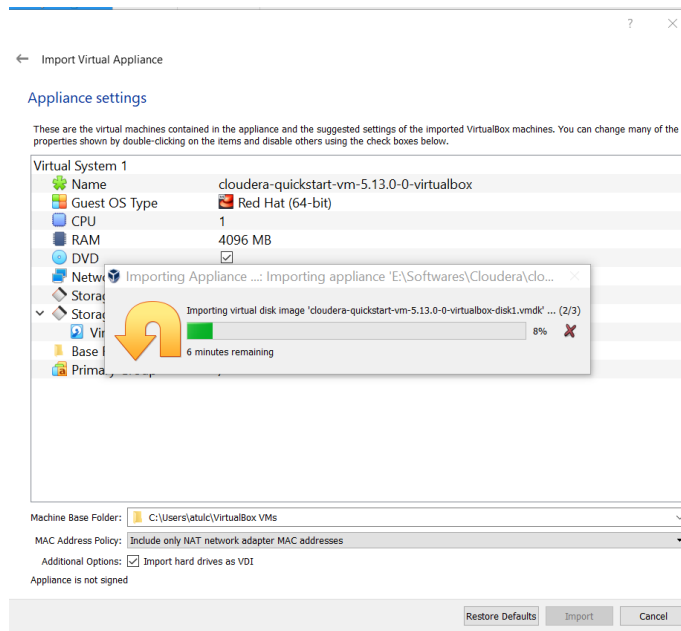
Step 3: Now browse your cloudera file and then click on next.



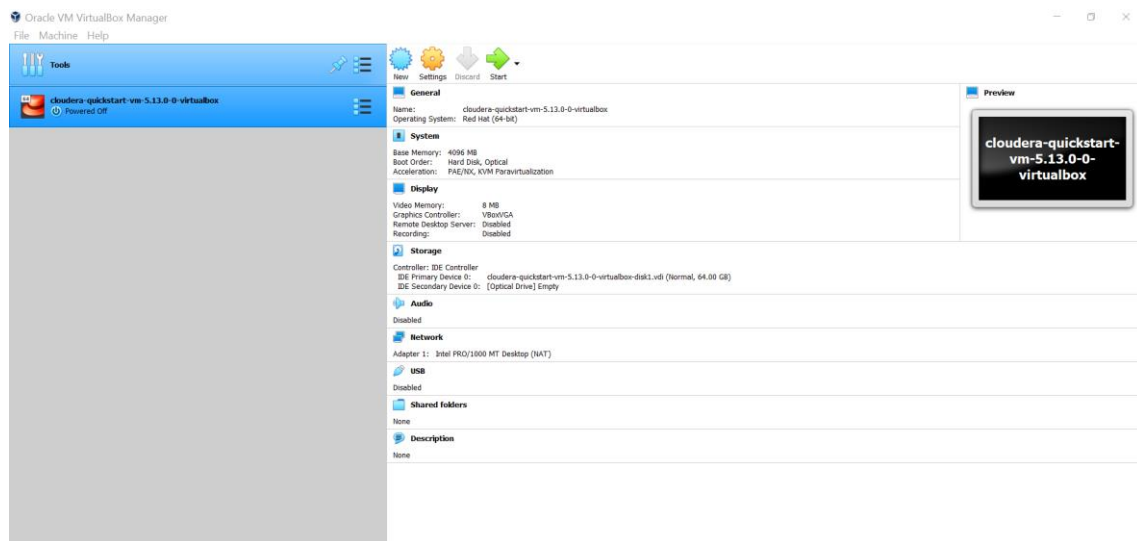
Step 4: After clicking on next now it shows all your cloudera appliance setting now click on Import.



Step 5: Now your Cloudera is importing you can also see the status of installation.

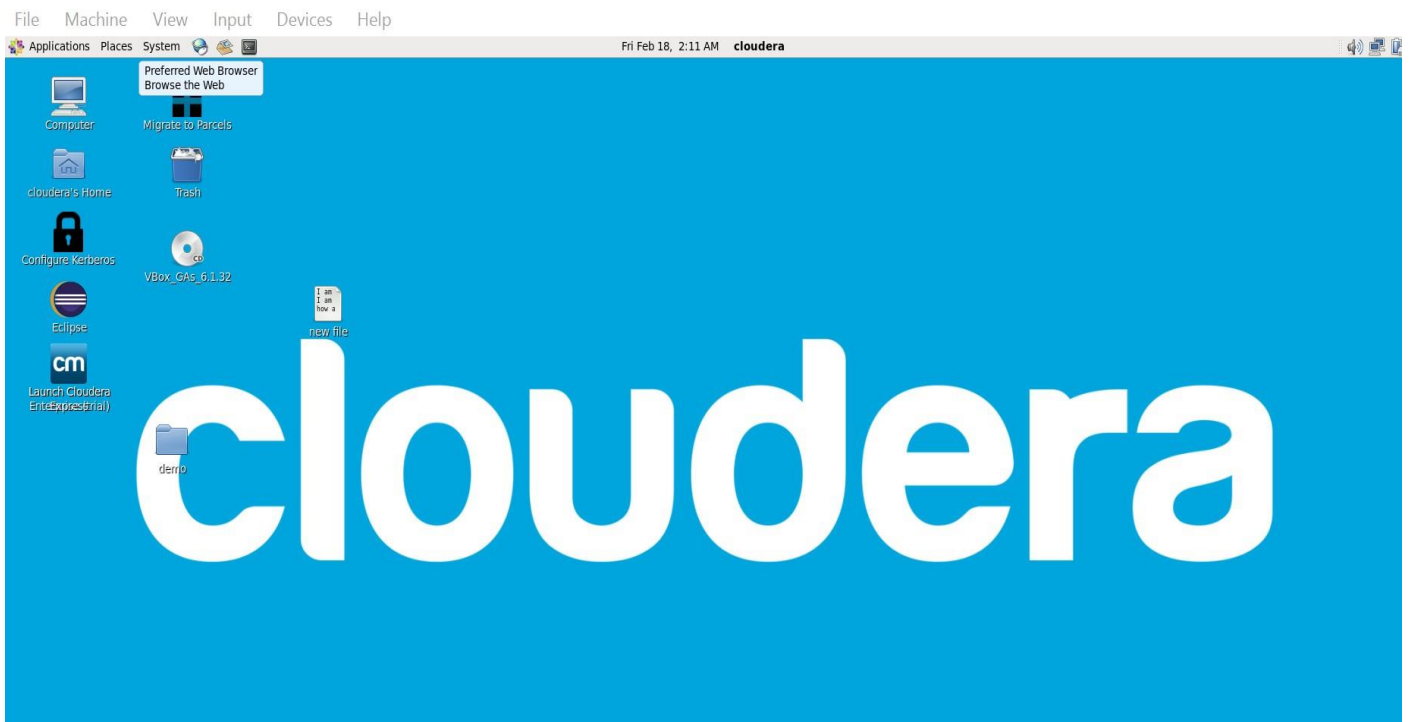


Step 6: Now your Cloudera QuickStart is installed and ready to use. You can start it by clicking on Start option or you can double tap on the machine.

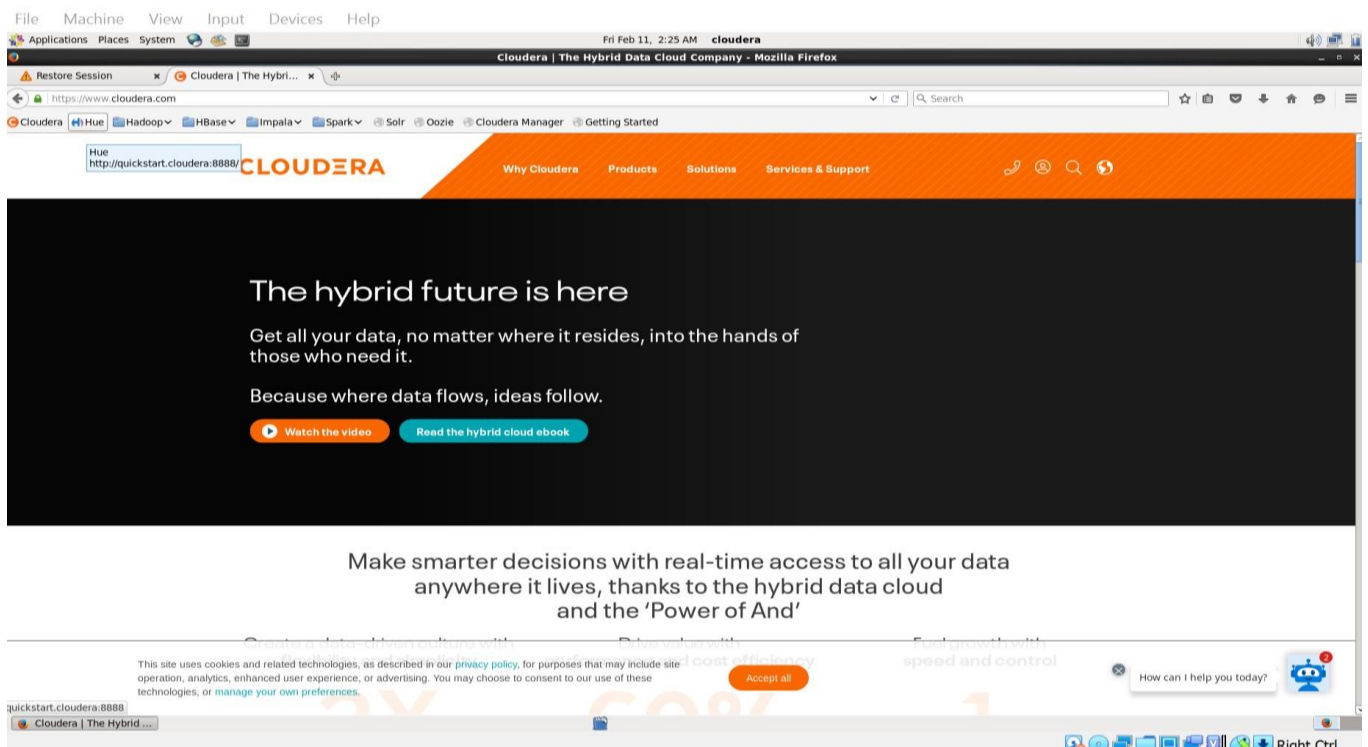


Task 4: CREATING TABLE USING HIVE EDITOR.

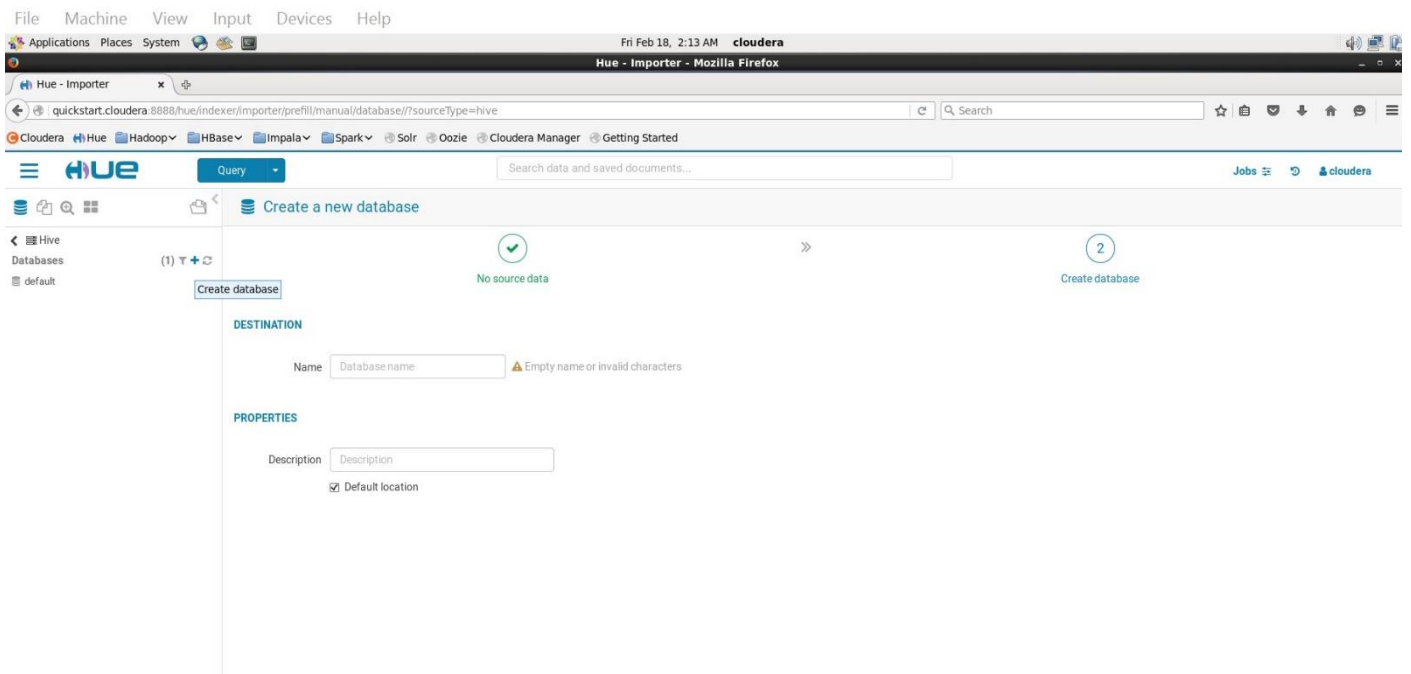
Step 1: First open the Cloudera QuickStart VM and then click on the web browser.



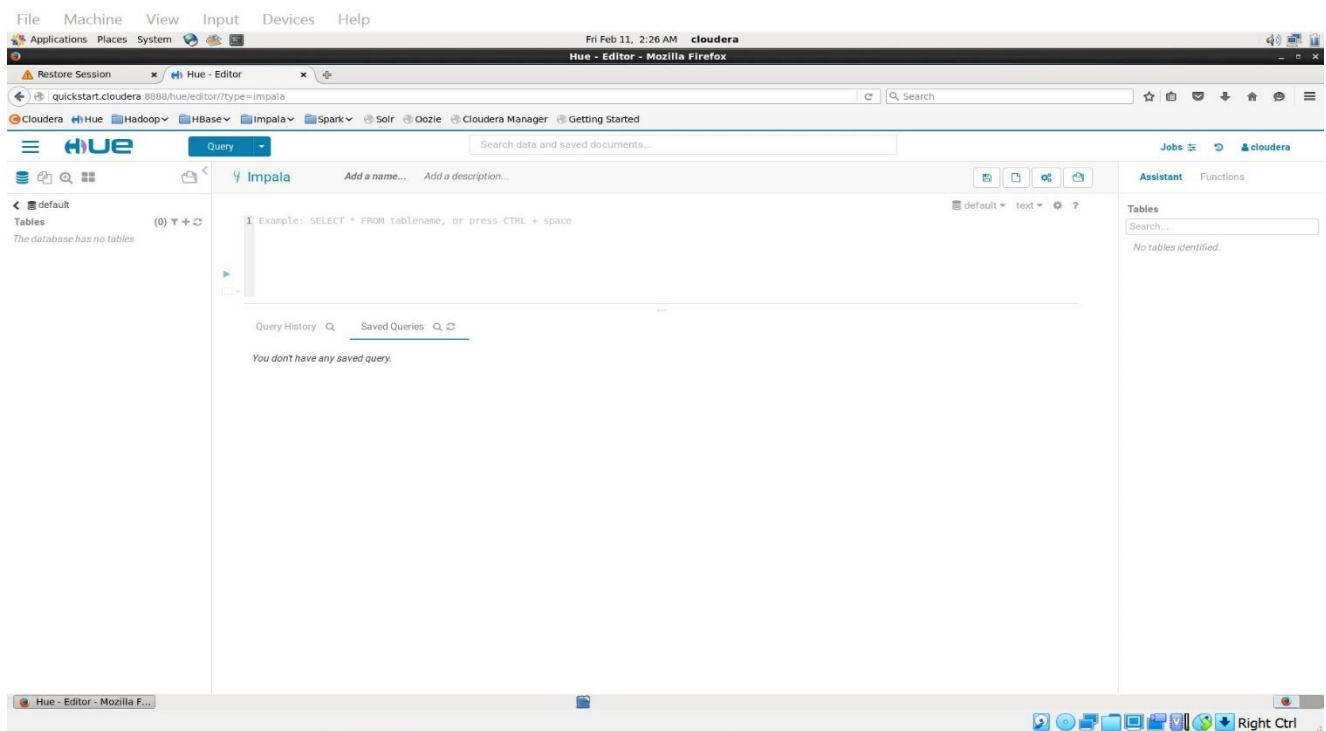
Step 2: Now click on the Hue option as shown in the screenshot.



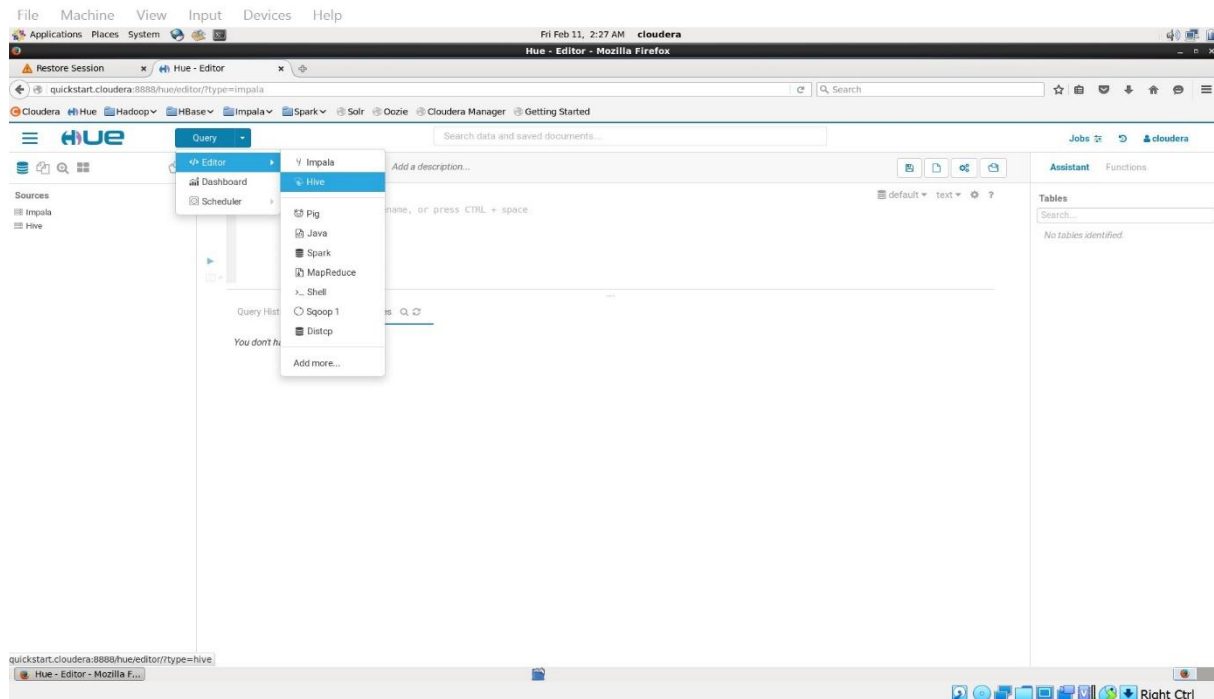
Step 3: For creating new database click on the '+' icon as shown in the screenshot.



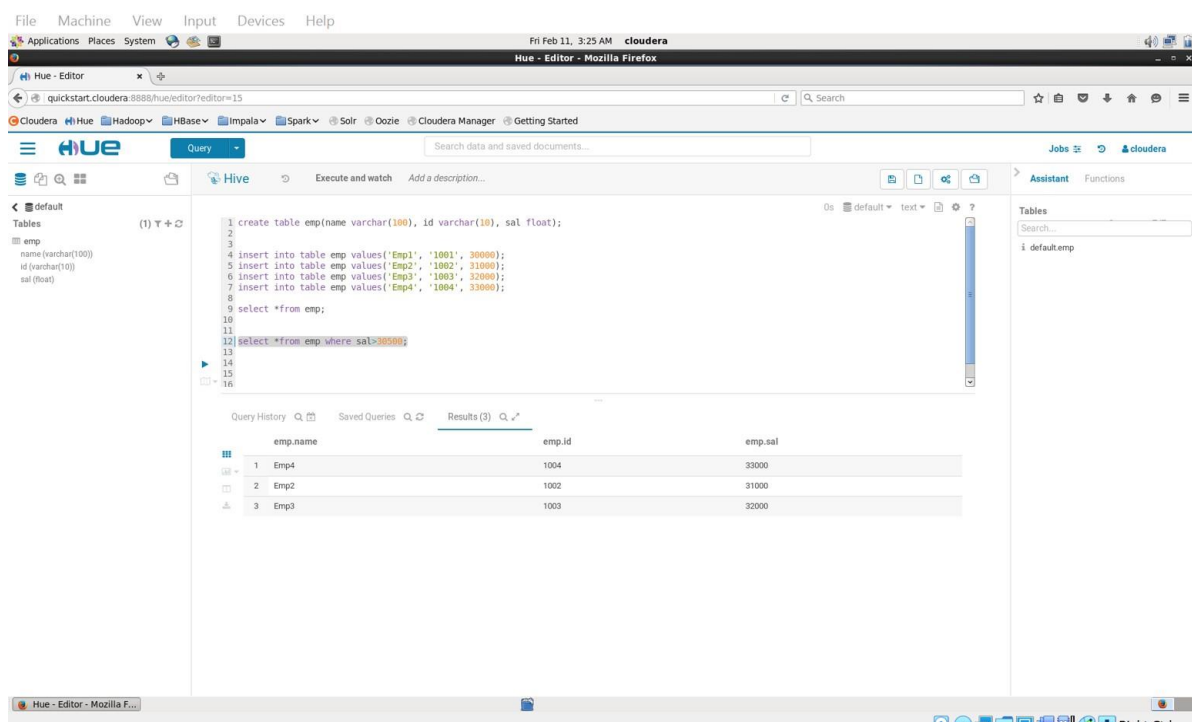
Step 4: Now, click on the down-arrow option that is just near the Query.



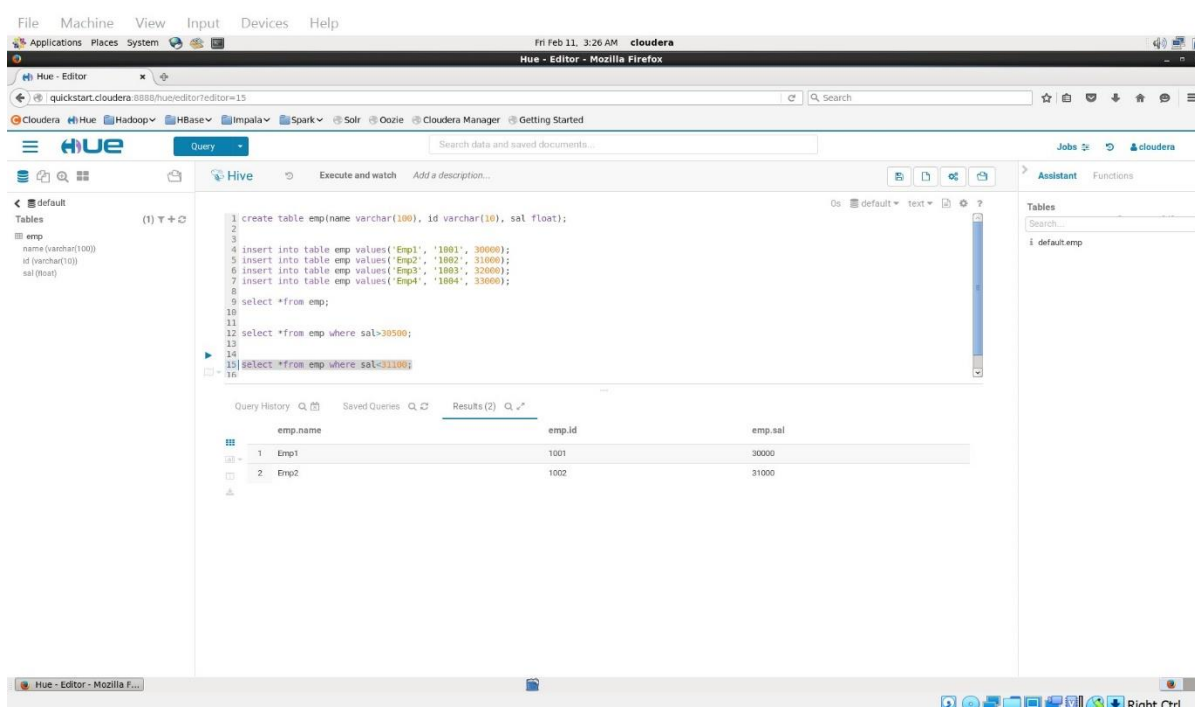
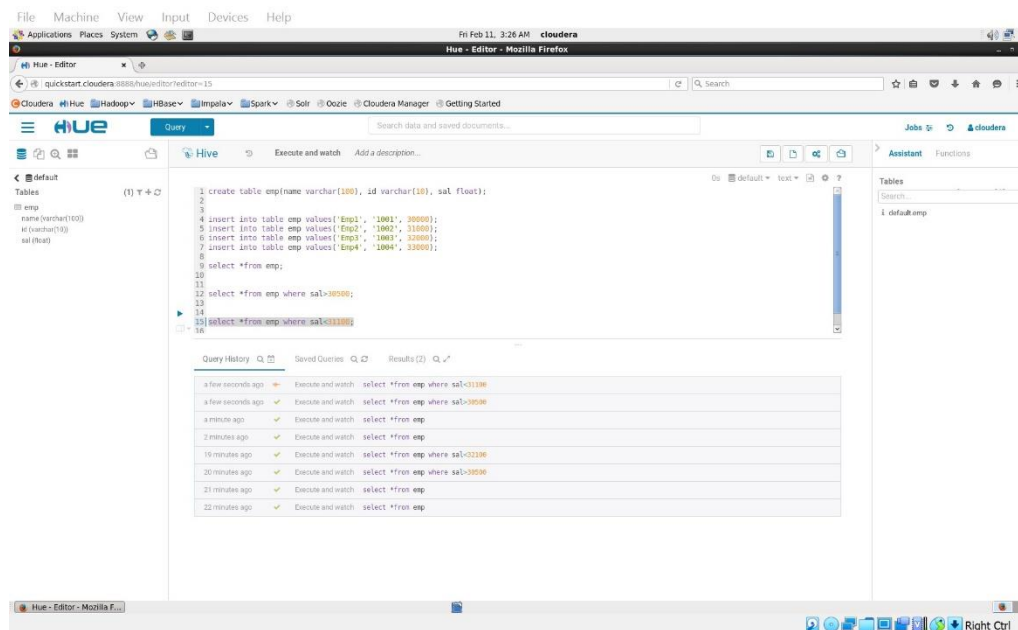
Step 5: Now, follow the path as shown in the screenshot and click on the Hive Editor.



Step 6: Here we create a table having name 'emp' and having attribute- name, id and sal.

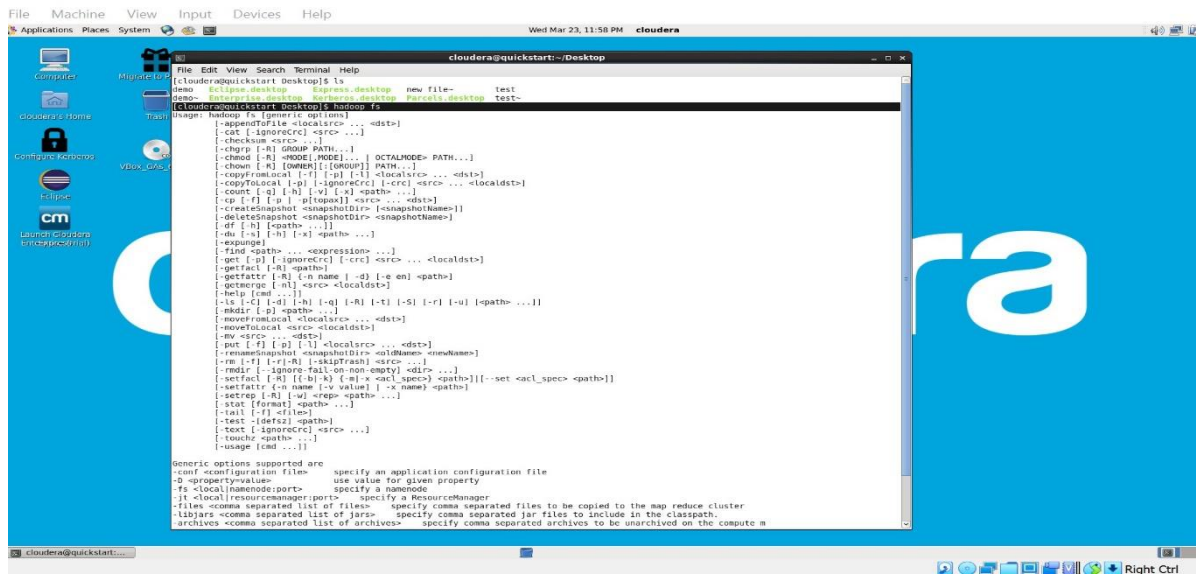


Step 7: Now in below screenshots, here we perform some simple query to extract the data from table.

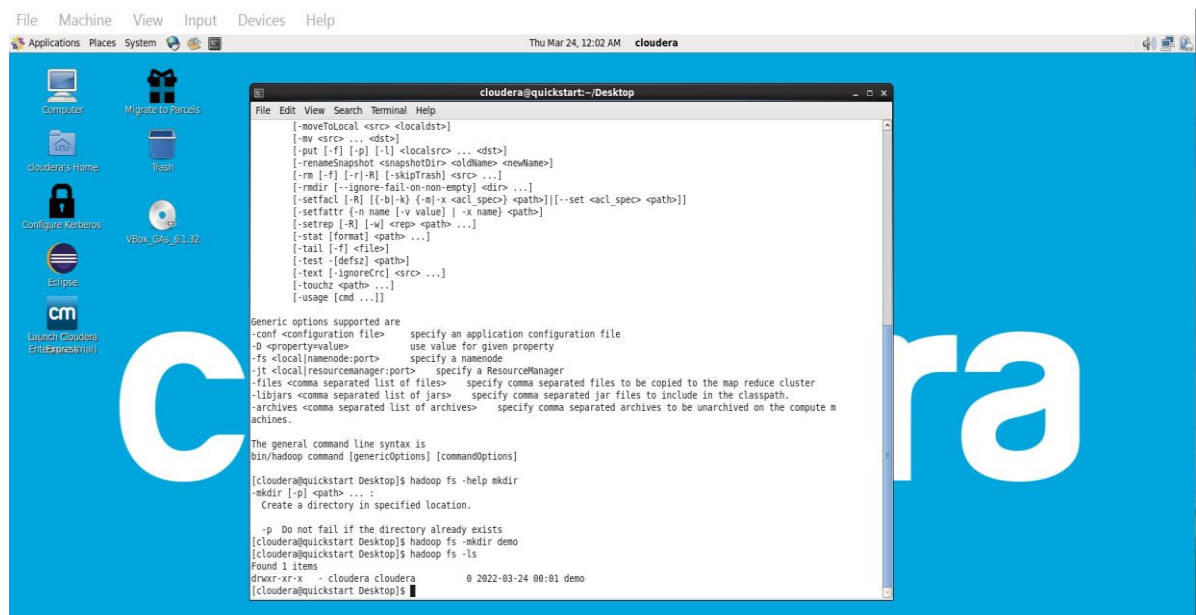


Task 5: INTERACTION WITH HADOOP FILE SYSTEM COMMANDS LIKE HADOOP FS, COPY A FILE FROM DESKTOP TO HADOOP & HADOOP TO DESKTOP, CHECK HOW TO RUN JAR, MAPREDUCE WORDCOUNT

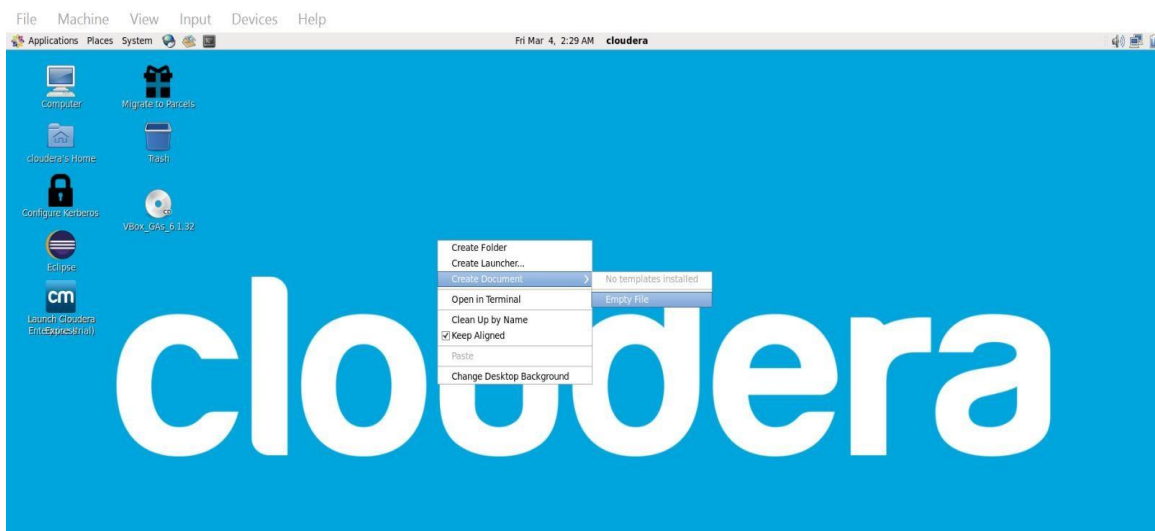
Step 1: “hadoop fs” command -

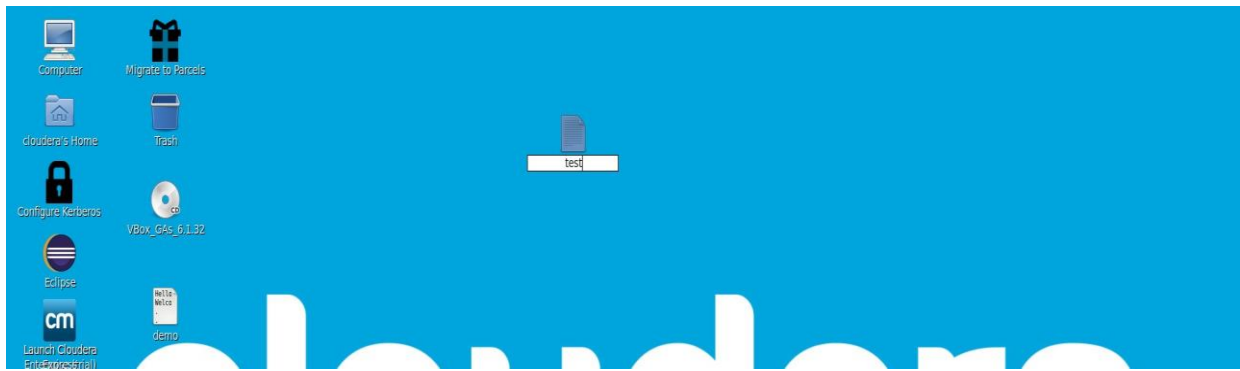


Step 2: Some basic commands -

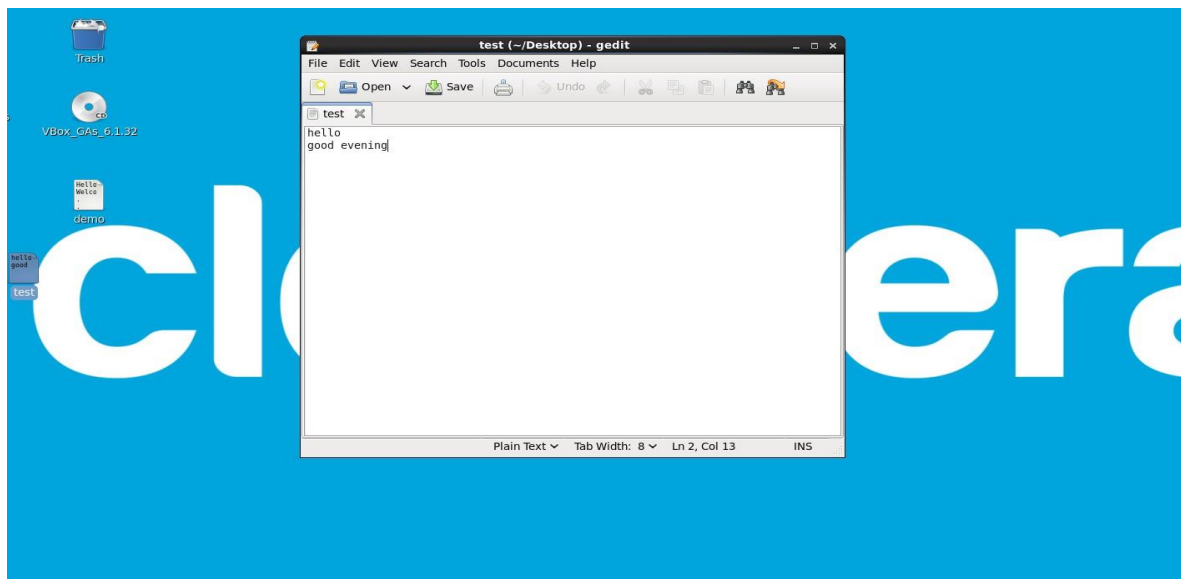


Step 3: First create a text file on Cloudera desktop as shown below in the screenshots.

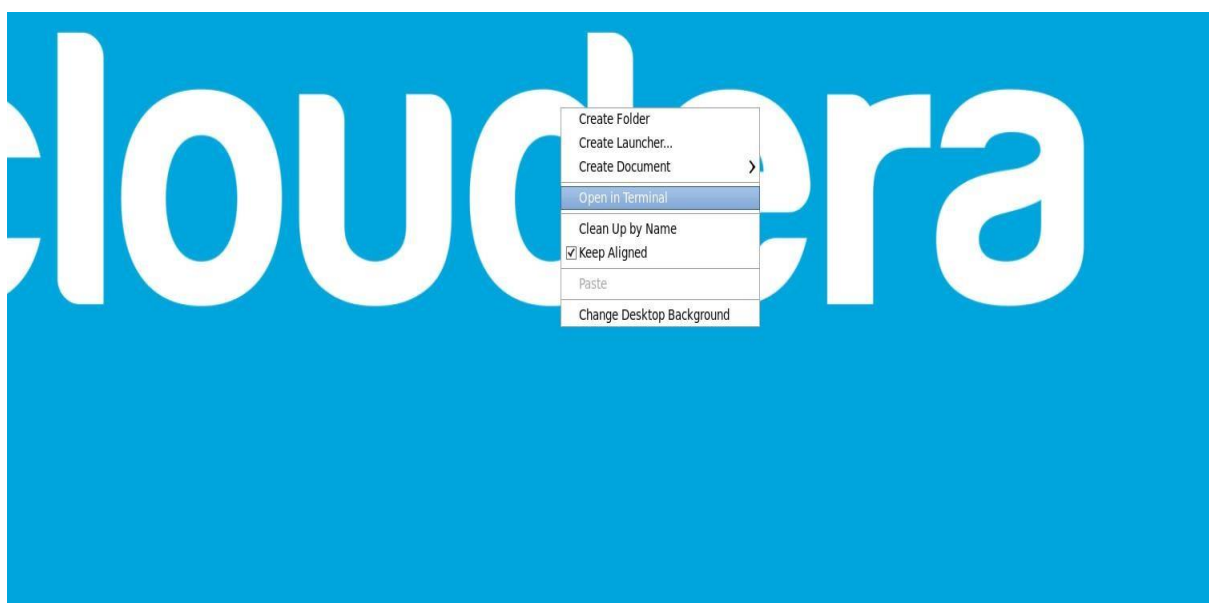




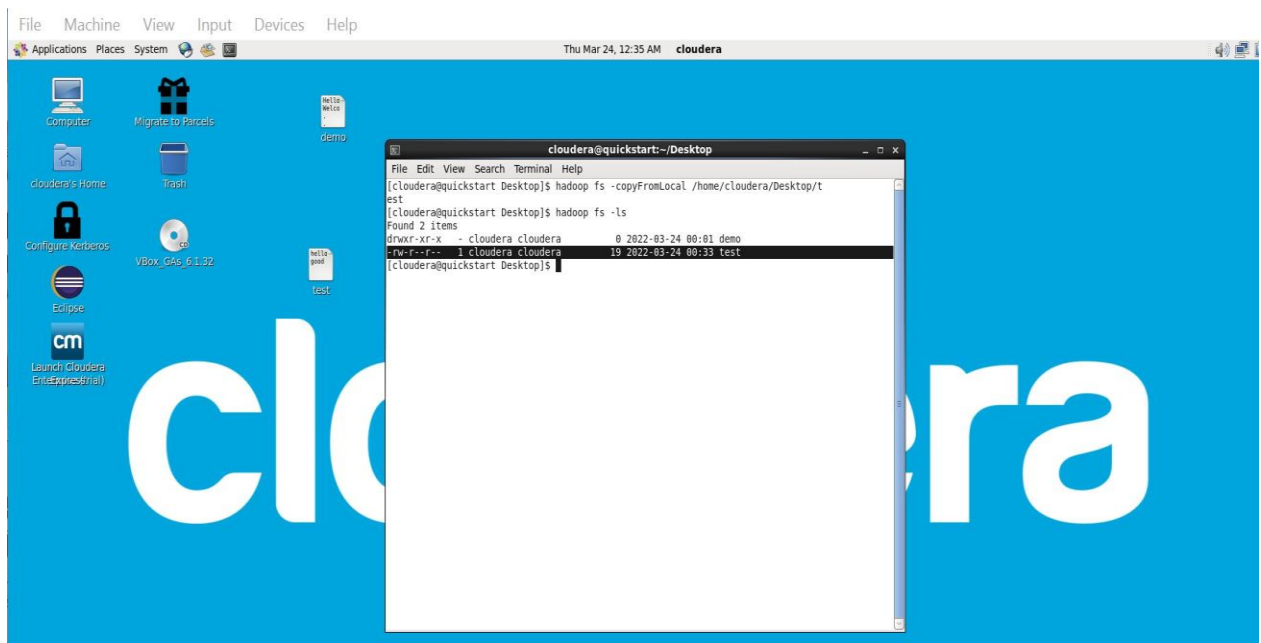
Step 4: Write something inside the text file and save it, by clicking on save button.



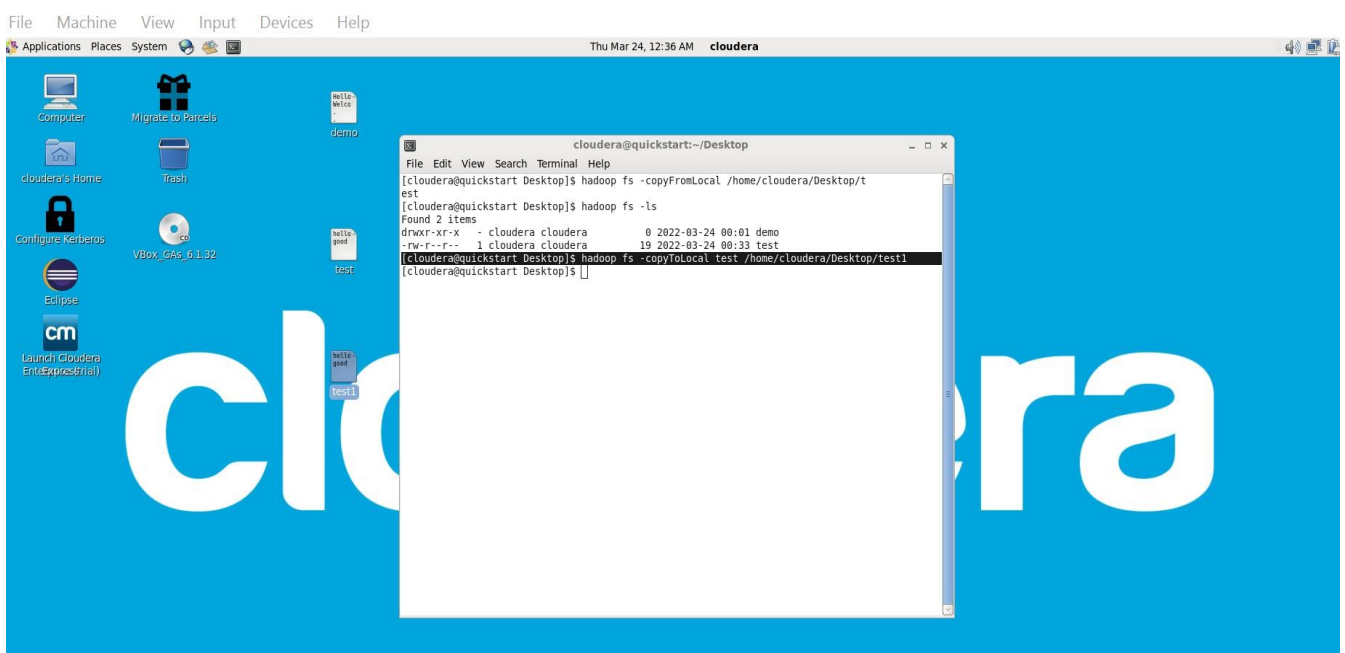
Step 5: Now open the terminal by clicking right mouse button and select open in terminal.

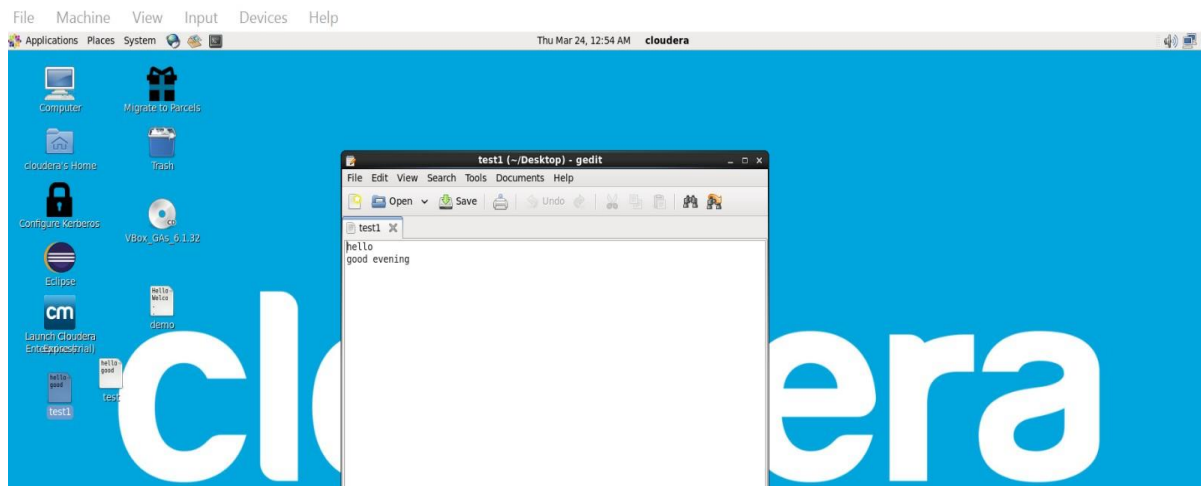


Step 6: copy a file from cloudera desktop (or another place) to hadoop fs



Step 7: copy a file/folder from hadoop fs to cloudera home/any location





Step 8: one jar file executed for MapReduce task for wordcount from a file. the MapReduce example was there in `usr/lib/hadoop-mapreduce-examples.jar`

